

WHEN $X \sim N(\mu, \sigma^2)$, WE HAVE $\bar{X} \perp S^2$

INDEPENDENT
SAMPLE VARIANCE

TO SEE THIS, IT IS ENOUGH TO CHECK THAT $\bar{X} \perp X_i - \bar{X}$, $\forall i$

$\Leftrightarrow \text{COV}(\bar{X}, X_i - \bar{X}) = 0$ SINCE X_i ARE NORMAL

S^2 IS A FUNCTION OF THESE TERMS

$$= \text{COV}(\bar{X}, X_i) - \text{COV}(\bar{X}, \bar{X}) = \text{COV}\left(\frac{1}{m} \sum_{j=1}^m X_j, X_i\right) - \frac{\sigma^2}{m}$$

$$= \frac{1}{m} \sum_{j=1}^m \text{COV}(X_j, X_i) - \frac{\sigma^2}{m} = \frac{1}{m} \text{COV}(X_i, X_i) - \frac{\sigma^2}{m} = 0$$

VARIANCE σ^2

= 0 WHEN $i \neq j$

EX: $X \sim N(10, 20)$. $m = 25$. THEN $\bar{X} \sim N(10, \frac{20}{25}) = N(10, 0.8)$

MEAN VARIANCE

$$(m-1) \frac{S^2}{\sigma^2} = \frac{24}{20} S^2 = 1.2 S^2 \sim \chi_{24}^2$$

DEGREE OF FREEDOM

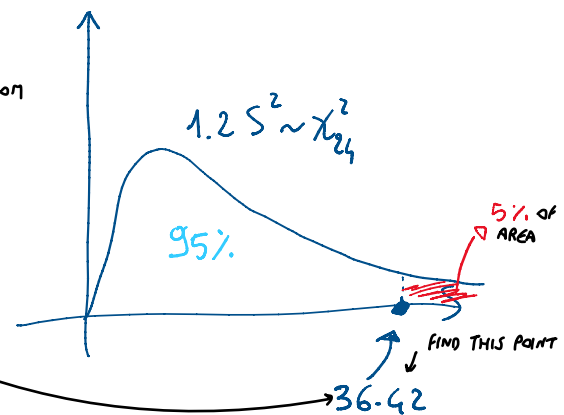
$$P(1.2 S^2 > x) = 0.05$$

AREA

$$\Rightarrow x = qchisq(0.95, df = 24) = 36.42$$

AREA OF THE LEFT
PROB. ON ITS LEFT

DEGREES OF FREEDOM



$$\Rightarrow P(1.2 S^2 > 36.42) = 0.05 \Rightarrow P(S^2 > 36.42 / 1.2) = 0.05$$

30.35

STANDARDIZED $\bar{X}$

STANDARDIZED SAMPLE MEAN
 $\hookrightarrow$ MEAN = 0; VARIANCE = 1

$$Z = \frac{\bar{X} - \mu}{\sqrt{\sigma^2/m}}$$

WHEN $X \sim N(\mu, \sigma^2) \Rightarrow Z \sim N(0, 1)$

LINEAR TRANSFORMATION OF NORMAL RANDOM VARIABLE ARE NORMAL

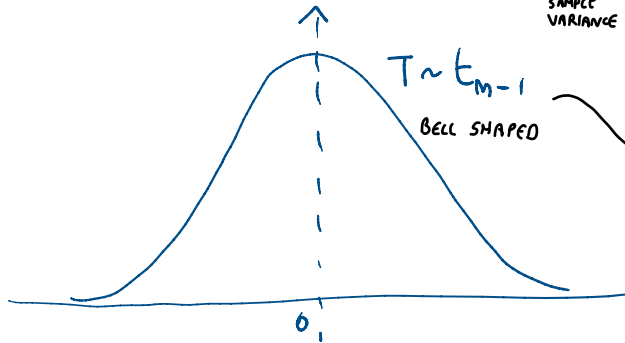
MEAN VARIANCE

USUALLY, σ^2 IS UNKNOWN $\Rightarrow$ $T = \frac{\bar{X} - \mu}{\sqrt{S^2/m}} \sim t_{m-1}$

INSTEAD OF Z USE T AND REPLACE σ^2 WITH S^2

T-STUDENT DISTRIBUTION WITH $m-1$ df

SAMPLE VARIANCE



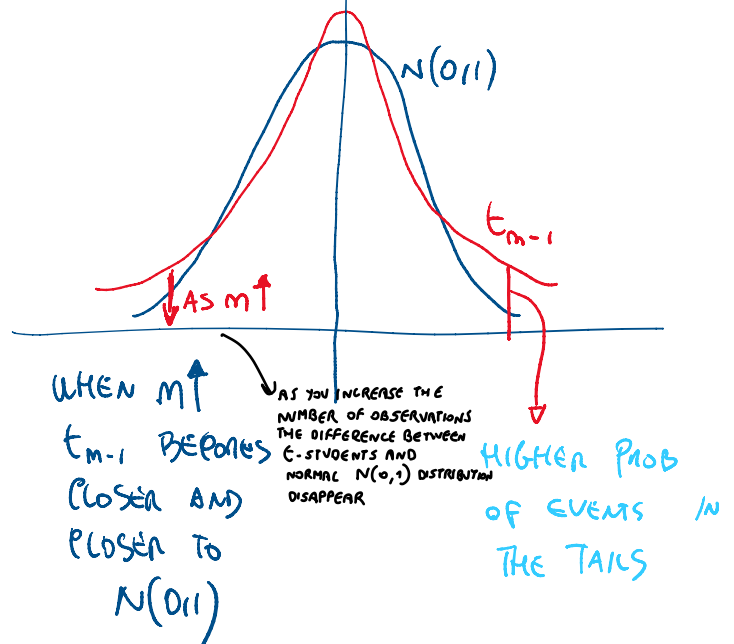
- SYMMETRIC
- HEAVIER TAILS THAN $N(0,1)$
 $\hookrightarrow$ EXTREME EVENTS IN THE TAIL HAVE AN HIGHER PROBABILITY TO BE REALIZED

IN GENERAL:

$$\frac{Z}{\sqrt{V/n}} \sim t_n \quad \text{WHEN } Z \sim N(0,1) \quad V \sim \chi_n^2$$

χ DEGREE OF FREEDOM

RELATIONSHIP BTW t-STUDENT AND $N(0,1)$, χ^2



CONFIDENCE INTERVAL $\rightarrow$ HYPOTHESIS TESTING

IN OUR CASE:

$$\frac{\bar{X} - \mu}{\sqrt{S^2/m}} = \frac{\bar{X} - \mu}{\sqrt{\frac{(m-1)S^2}{m}}} \cdot \frac{\sqrt{m}}{\sqrt{m-1}}$$

MULTIPLY AND DIVIDE BY SAME QUANTITIES

χ_{m-1}^2

$\frac{\bar{X} - \mu}{\sqrt{\sigma^2/m}} \sim N(0,1)$

χ_{m-1}^2

TO FIND QUANTILES OF t-STUD:

qt (IN R)

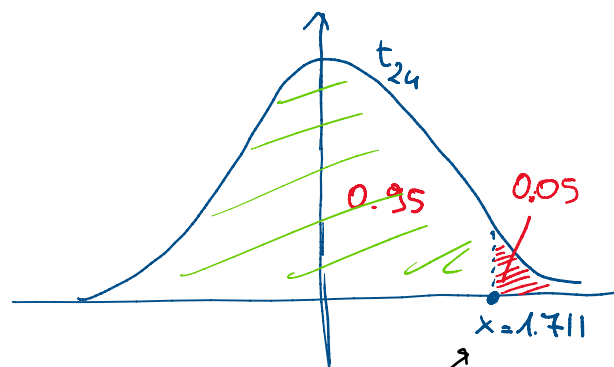
scipy.stats.t.ppf (PYTHON)

tinv (IN MATLAB/EXCEL)

THESE QUANTILES WILL BE USEFUL FOR CI AND HT FOR μ WHEN σ^2 UNKNOWN

EX: $X \sim N(10, 20)$, $n=25$

$$T = \frac{\bar{X} - \mu}{\sqrt{S^2/n}} = \frac{\bar{X} - 10}{\sqrt{S^2/25}} \sim t_{24}$$



WHAT WE WANT TO FIND

$$P(T > x) = 0.05$$

$$\Rightarrow P(T < x) = 0.95$$

$$\Rightarrow F_T(x) = 0.95 \quad \text{WHERE } F_T \text{ IS CDF OF } T$$

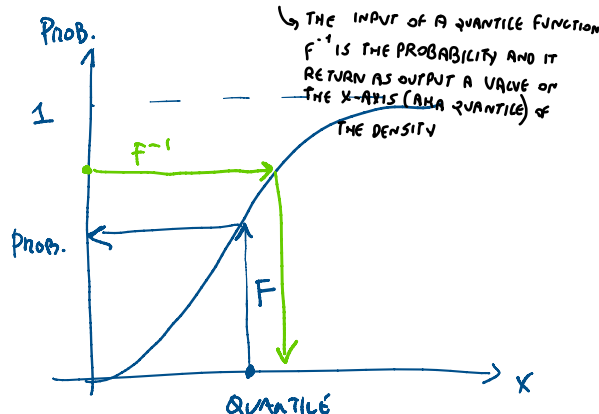
$$\Rightarrow x = F_T^{-1}(0.95) \quad \text{WHERE } F_T^{-1} \text{ IS THE INVERSE CDF OF } T$$

$$F_T^{-1}(F_T(x))$$

DEGREE OF FREEDOM

$$x = qt(0.95, 24) = 1.711$$

QUANTILE FUNCTION



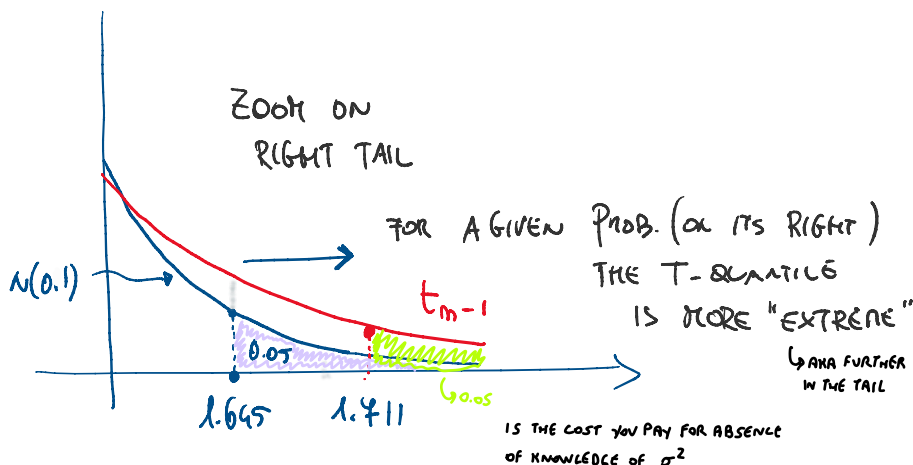
VS A CORRESPONDING QUANTILE FOR $N(0,1)$ WHICH IS

$$qnorm(0.95) = 1.645$$

1.645

1.711

$qt > qnorm$ BECAUSE OF HEAVIER TAILS OF t -STUD:



IF WE INCREASE $n = 101$

$$\hookrightarrow T = \frac{\bar{X} - \mu}{\sqrt{S^2/101}} \sim t_{100}$$

CONSIDERED A BIG DEGREE OF FREEDOM VALUE

$$qt(0.95, 100) = 1.66$$

MUCH CLOSER TO 1.645 OF $N(0,1)$

SUFFICIENCY

NOTES 1.7

WANT TO REDUCE THE COMPLEXITY OF THE DATA WITHOUT LOSING IMPORTANT INFORMATION

STATISTIC $Y = t(X)$ CREATES A PARTITION OF THE SAMPLE SPACE Ω (X BEFORE)

INTO SUBSETS $\Omega_y = \{x \in \Omega : t(x) = y\}$. IF I COMPUTE THE SAMPLE STATISTIC ON THE DATA I ALWAYS OBTAIN Y

WITH THE CHOICE OF $Y = t(X)$, I DECIDE TO NEGLECT DIFFERENT SAMPLES THAT PROVIDE THE SAME VALUE OF THE STATISTIC

$\Rightarrow$ IF $\overset{\text{SAMPLES}}{x_1}$ AND x_2 ARE SUCH THAT $t(x_1) = t(x_2)$, I WILL DRAW THE SAME INFERENCE CONCLUSIONS ON θ

ANY STATISTIC INDUCES A DATA REDUCTION (DATA SUMMARY)

STILL YOU DO NOT WANT TO LOSE IMPORTANT INFO. ON θ

SUFFICIENT STATISTIC: STATISTIC THAT SUMMARIZES DATA AND DO NOT LOSE INFO ON θ (THETA)

FORMALLY: SUFFICIENT IF "CONDITIONAL DISTRIB. OF $\overset{\text{SAMPLE}}{X}$ GIVEN $Y = t(X)$ DOES NOT DEPEND ON θ "

$$f(x | Y = t(x); \theta) = \frac{\overset{\text{JOINT DISTRIBUTION OF THE SAMPLE}}{f(x; \theta)}}{\underset{\text{DISTRIBUTION OF THE STATISTIC}}{f(t(x); \theta)}} \quad \text{CONSTANT AS A FUNCTION OF } \theta \Rightarrow t(x) \text{ SUFFICIENT}$$

IF THE CONDITIONAL DISTRIBUTION DOES NOT DEPEND ON θ THE STATISTIC $t(x)$ IS SUFFICIENT

NOTE: NOTATION IS $f(x; \theta)$ AND NOT $f(x, t(x); \theta)$

SINCE $f(t(x) | x; \theta) \cdot f(x; \theta)$ SINCE IF YOU KNOW x YOU ALREADY KNOW $t(x)$

= 1

BECAUSE IF YOU KNOW THE WHOLE SAMPLE I ALSO KNOW THE VALUE OF THE STATISTIC

EX: $X \sim \text{Ber}(p)$. I CHOOSE AS STATISTIC "NUM SUCCESSES"

I WANT TO FIND THE NUMERATOR AND DENOMINATOR OF THAT CONDITIONAL PROBABILITY MASS FUNCTION (PMF)

$$Y = t(X) = \sum_{i=1}^m X_i$$

NUMERATOR

$$f(X; p) = \prod_{i=1}^m f(X_i; p) = \prod_{i=1}^m p^{X_i} (1-p)^{1-X_i} = p^{\sum_{i=1}^m X_i} (1-p)^{m - \sum_{i=1}^m X_i}$$

$$= p^Y (1-p)^{m-Y} \quad (T=t(X)=Y)$$

$$f(\sum_{i=1}^m X_i; p) = f(Y; p) = \binom{m}{Y} p^Y (1-p)^{m-Y} \quad (\text{BINOMIAL})$$

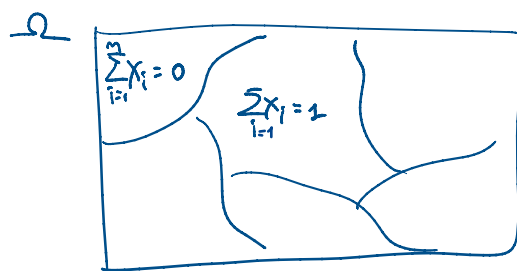
$$f(X | \sum_{i=1}^m X_i = Y; p) = \frac{f(X; p)}{f(\sum_{i=1}^m X_i = Y; p)} = \frac{p^Y (1-p)^{m-Y}}{\binom{m}{Y} p^Y (1-p)^{m-Y}} = \frac{1}{\binom{m}{Y}}$$

DOES NOT DEPEND ON $p \Rightarrow \sum_{i=1}^m X_i$ SUFFICIENT FOR p

$$Y \sim \text{Bin}(m, p) \Rightarrow \text{Supp}(Y) = \{0, 1, \dots, m\}$$

WHEN WE COLLECT DATA I CHOOSE TO JUST LOOK AT THE SUM OF AXIS. IF THE SUM ARE THE SAME I DO NOT DISTINGUISH BETWEEN THESE TWO SAMPLES

$\Rightarrow Y$ CREATES A PARTITION OF Ω OF SIZE $m+1$



EX:

$X \sim N(\mu, \sigma^2)$ σ^2 KNOWN, μ UNKNOWN

CHECK THAT THE CONDITIONAL DISTRIBUTION OF THE WHOLE SAMPLE GIVEN $\bar{X}$ DOES NOT DEPENDS ON μ

WE CHOOSE $Y = t(X) = \bar{X}$

TO CHECK THAT THE USE OF $\bar{X}$ DOES NOT WASTE INFO. ON μ , WE NEED $f(X | \bar{X} = y; \mu)$ FREE FROM μ .

$$= \frac{f(X; \mu)}{f(\bar{X}; \mu)}$$

$$f(\underline{x}; \mu) = \prod_{i=1}^m f(x_i; \mu) = \prod_{i=1}^m \frac{1}{\sqrt{2\pi} \sigma} \exp \left\{ -\frac{1}{2\sigma^2} (x_i - \mu)^2 \right\}$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \mu)^2 \right\}$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \bar{x} + \bar{x} - \mu)^2 \right\}$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^m [(x_i - \bar{x})^2 + (\bar{x} - \mu)^2 + 2(\bar{x} - \mu)(x_i - \bar{x})] \right\}$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^m (x_i - \bar{x})^2 + \underbrace{\sum_{i=1}^m (\bar{x} - \mu)^2}_{m(\bar{x} - \mu)^2} + 2(\bar{x} - \mu) \underbrace{\sum_{i=1}^m (x_i - \bar{x})}_{\sum x_i - \sum \bar{x}} \right) \right\}$$

$$t(\underline{x}) = \bar{x} = y$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^m (x_i - \bar{x})^2 + m(\bar{x} - \mu)^2 \right) \right\}$$

$$= \frac{1}{(\sqrt{2\pi} \sigma)^m} \exp \left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^m (x_i - y)^2 + m(y - \mu)^2 \right) \right\}$$

$$\begin{aligned} &= \sum_{i=1}^m x_i - m\bar{x} \\ &= \sum_{i=1}^m x_i - m \cdot \frac{1}{m} \sum_{i=1}^m x_i \\ &= 0 \end{aligned}$$

I know $\bar{x} \sim N(\mu, \sigma^2/m)$

$$\hookrightarrow f(\bar{x}=y; \mu) = \frac{1}{\sqrt{2\pi} \sigma/\sqrt{m}} \exp \left\{ -\frac{1}{2\sigma^2/m} (y - \mu)^2 \right\}$$

$$= \frac{\sqrt{m}}{\sqrt{2\pi} \sigma} \exp \left\{ -\frac{m}{2\sigma^2} (y - \mu)^2 \right\}$$

WILL CANCEL IN THE RATIO

$$f(\underline{x} | \bar{x} = y; \mu) = \frac{f(\underline{x}; \mu)}{f(\bar{x} = y; \mu)} = \frac{1}{(\sqrt{m}\sigma)^m} \exp\left\{-\frac{1}{2\sigma^2} \sum_i (x_i - y)^2\right\} \cdot \frac{\sqrt{m}\sigma}{\sqrt{m}}$$

$$= \frac{1}{\sqrt{m}(\sqrt{m}\sigma)^{m-1}} \exp\left\{ \dots \right\}$$

FREE FROM μ ☺

⇒ $\bar{X}$ SUFFICIENT FOR μ

There is a more automatic way to identify
SUFFICIENT STATISTICS ⇒ FACTORIZATION THEOREM :

$Y = t(\underline{X})$ IS SUFFICIENT FOR θ IFF

$$f(\underline{x}; \theta) = \underset{\substack{\uparrow \\ \text{DEPENDS ON STATISTIC}}}{g(t(\underline{x}); \theta)} \cdot \underset{\substack{\uparrow \\ \text{DEPENDS ONLY ON } \underline{x}}}{h(\underline{x})}, \quad \forall \underline{x}, \theta$$

($h(\underline{x})$ DOES NOT DEPEND ON θ)

ANY TRANSFORMATION OF A
SUFFICIENT STATISTIC IS AGAIN
SUFFICIENT

Implication: ANY ONE-TO-ONE FUNCTION OF A SUFFICIENT
STATISTIC $t(\underline{x})$ IS ALSO SUFFICIENT

$$\Rightarrow S(\underline{x}) = u(t(\underline{x}))$$

$$\Rightarrow \underline{t(\underline{x}) = u^{-1}(S(\underline{x}))}$$

$$\Rightarrow f(\underline{x}; \theta) = g(t(\underline{x}); \theta) h(\underline{x}) \quad \left(\begin{array}{l} \text{DOABLE SINCE} \\ t(\underline{x}) \text{ SUFF.} \end{array} \right)$$

$$= g(\underline{u^{-1}(S(\underline{x}))}; \theta) h(\underline{x})$$

$$= \tilde{g}(S(\underline{x}); \theta) \cdot h(\underline{x})$$

IF ONE STATISTIC IS SUFFICIENT
ANY ONE-TO-ONE FUNCTION OF
IT IS ALSO SUFFICIENT

$$\text{WHERE } \tilde{g}(S(\underline{x}); \theta) := g(u^{-1}(t(\underline{x})); \theta)$$

WITHIN EXPONENTIAL FAMILY: $f(x; \theta) = h(x) d(\theta) \exp \left\{ \underbrace{\sum_{j=1}^k w_j(\theta) t_j(x)}_{g(t(x); \theta)} \right\}$

$\Rightarrow$ FROM FACTORIZ. TH, $Y = (t_1(x), t_2(x), \dots, t_k(x))$ IS
SUFFICIENT FOR θ

EX: $X \sim \text{Bern}(p)$, X OF SIZE m $\overset{\text{INDI. ID. DISTRIBUTED}}{\parallel}$

$$\begin{aligned} f(x; p) &= p^{\sum_i x_i} (1-p)^{m - \sum_i x_i} \\ &= \underbrace{p^{\sum_i x_i} (1-p)^{m - \sum_i x_i}}_{g(t(x); p)} \cdot \underbrace{1}_{h(x)} \quad \Rightarrow \sum_i x_i \text{ SUFF. FOR } p \end{aligned}$$

$\Rightarrow$ ALSO $\bar{X}$ IS SUFF. FOR p , SINCE ONE-TO-ONE FUNCTION OF $\sum_i x_i$

⌊ (WE ALREADY KNEW THAT $\sum_i x_i$ SUFF. SINCE Bern. $\in$ exp. family)

Typically: MANY SUFFICIENT STATISTICS FOR θ

- $t(x) = X$ ^{IF YOU DO NOT REDUCE YOUR DATA YOUR DATA WILL BE SUFFICIENT} (TRIVIAALLY SUFFICIENT FOR θ , WITH NO DATA REDUCTION)
↳ FINEST PARTITION OF Ω
- ANY ONE-TO-ONE ^{FUNCTION} FCN OF A SUFFICIENT STATISTIC IS SUFFICIENT
(THEY GIVE THE SAME PARTITION OF Ω)
↑
SAMPLE SPACE

$\Rightarrow$ DIFFERENT SUFF. STATISTICS LEAD TO DIFFERENT PARTITIONS OF Ω

$\Rightarrow$ I WOULD LIKE TO CHOOSE THAT SUFF. STAT. WITH
COARSEST PARTITION